Orbital Contact Analyzer — User Manual

Version: v1.4.0 **Scope:** Single-file, offline HTML app for simulating two Earth satellites and up to two ground stations. The application uses a J2 perturbation model, renders 2D/3D views, visualizes sensor footprints and local horizons, and computes both satellite-to-satellite line-of-sight (LOS) and satellite-to-ground-station access intervals.

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1) Overview

This app propagates two satellites using a model that includes J2 secular perturbations, converts positions between ECI and ECEF frames, and visualizes:

- **2D Map:** Equirectangular projection with ground tracks, current markers, sensor footprints (FOV), local horizons, and communication links.
- **3D Globe:** WebGL-based view with full camera control, physically correct inertial and Earth-fixed views, and a full-screen “Focus Mode.”
- **Scenario Loader:** A quick-configuration menu to instantly load common orbital regimes (LEO, GEO, Molniya, etc.).
- **Data Tables:** Lists for Inter-Satellite LOS and Ground Station Access intervals, with a single-click CSV export.

Key design choices:

- **J2 Perturbed propagation:** Includes secular effects on RAAN (Ω) and Argument of Perigee (ω).
- **Spherical Earth model:** Used for all ground geometry, occlusion checks, and footprint calculations.
- **Epoch anchoring:** The simulation timeline starts at a user-defined UTC epoch. Each satellite additionally carries its own element anchor: hand-entered elements are valid at the simulation epoch, while TLE-imported elements are anchored at the TLE's own epoch and propagated onto the timeline.
- **Sub-second contact boundaries:** Access interval start and end times are refined by bisection between time samples rather than being quantized to

the step size. This solves the *model's* access transitions to about 1 ms; real-world accuracy is limited by the model physics, not the grid (see Simulation Fidelity & Limitations).

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2) System Requirements

- A modern browser with **WebGL enabled** (Chrome, Edge, Firefox, Safari), on desktop, tablet, or phone.
- **Touch is fully supported:** one finger rotates the 3D globe, two fingers pinch-zoom, and the layout adapts to narrow screens.
- 8+ GB RAM recommended for long timelines or small time steps.
- **Runs fully offline;** no external network access is required.

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3) Concepts in Brief

- **Orbital Elements (per satellite):**
  - a (km), e, i (deg), Ω (RAAN), ω (ArgP), M (Mean Anomaly).
  - **Min. elevation ϵ (deg):** Defines the sensor footprint (Field of View).
- **Ground Stations:** Up to two points on the Earth's surface, each defined by Latitude, Longitude, and a Min. Elevation Mask (deg) used to calculate access. Ground Station 1 (GS1) is shown as a white triangle; Ground Station 2 (GS2) is shown as a green triangle.
- **J2 Perturbations:** The application models the primary secular (long-term) effects of Earth's equatorial bulge, which causes the orbital plane (Ω) and the orbit's orientation within that plane (ω) to precess over time.
- **Frames:**
  - **ECI** (Earth-Centered Inertial) for orbital dynamics.
  - **ECEF** (Earth-Centered Earth-Fixed) for surface overlays.
- **Inter-Satellite LOS:** A direct line-of-sight exists if the straight line connecting the two satellites is not obstructed by the spherical Earth.

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4) Quick Start

1. **Open the App:** Open the `index.html` file in your browser.

2. **Load a Scenario (Recommended):** At the top of the **Simulation Controls** panel, use the **Load Scenario** dropdown to instantly configure the simulation.
    - *Select "LEO Formation":* To see two satellites in low Earth orbit passing over a ground station.
    - *Select "Sun-Synchronous":* To demonstrate J2 perturbation effects over a 24-hour period.
    - *Select "Molniya":* To view high-eccentricity orbits with long dwell times over the northern hemisphere.
3. **Animate:** Click the green ▶ **Run** button to start the time evolution.
    - Use the **Scrub Bar** (bottom of the view) to manually drag time forward or backward.
    - *Note:* If you manually change any input number (e.g., Altitude), the scenario dropdown will automatically switch to **"Custom"**, preserving your edits.
4. **Visualize:**
    - Use the **Projection** dropdown (top right) to switch between the **2D Map** and **3D Globe**.
    - Use **Trail (orbits)** beside it to shorten long ground tracks: enter a number of orbits to draw as a trailing window, or leave it blank for the full time span.
    - Click **Focus Mode** (`↗`) to expand the visualization to fill your screen.
5. **Analyze Data:** Scroll down to view the **Line-of-Sight** and **Ground Station Access** tables. Click **Export CSV** to download a spreadsheet of all computed windows.

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5) Controls Reference

5.1 Simulation Controls (Top-Left)

- **Load Scenario:** A preset menu that configures the entire application state (Satellites, Ground Stations, and Time settings) for specific astrodynamics demonstrations.
- **Epoch (UTC, ISO-8601):** Global start time (e.g., `2025-11-18T12:00:00Z`).
- **Step Δt (s):** Simulation step size. Smaller is more precise but computationally heavier.
- **Duration (min):** Total span simulated from the epoch.

- **Speed (steps/frame):** Simulation steps advanced per animation frame while running. The equivalent real-time multiple depends on the step size and your display refresh rate.
- **Buttons:**
  - ▶ **Run / Pause:** Start or stop time stepping.
  - ↺ **Reset:** Rewind to the epoch.
  - **Recompute tracks:** Re-runs the simulation with current inputs (useful if you change an orbit manually).
  - **Export CSV:** Saves all calculated contact windows to a file. Timestamps carry millisecond precision and durations are displayed at 0.1 s resolution; both reflect the resolution of the model's solution, not real-world accuracy.

5.2 Satellite Panels

- **Elements ($a, e, i, \ldots$):** Standard Keplerian inputs.
- **Min. elevation ϵ:** Defines the size of the dashed FOV circle.
- **Visibility checkboxes:** Show Satellite, Show Track, Show FOV, and Show Horizon are one-tap checkboxes. The sub-toggles gray out while the satellite itself is hidden.
- **Note:** hand-editing any orbital element clears that satellite's TLE epoch anchor (typed elements are taken as valid at the simulation epoch; see 5.6).

5.3 Ground Station Panels

The application supports two independent ground stations. Ground Station 1 (GS1) is enabled by default; Ground Station 2 (GS2) is disabled by default and can be enabled when needed.

Each ground station panel contains:

- **Show Ground Station:** A checkbox that toggles visibility and enables access calculations for that station.
- **Lat/Lon:** Station coordinates.
- **Min Elev. Mask:** The minimum elevation angle (above the local horizon) required for a satellite to be considered "in view" from that station.

On the 2D map and 3D globe, GS1 appears as a **white** triangle and GS2 appears as a **green** triangle. When a satellite has access to a ground station, a dashed link line is drawn between them in the satellite's color.

The **Ground Station Access Intervals** table (below the map) merges intervals from both stations in chronological order. Each row indicates which station (GS1 or GS2) and which satellite are involved. The footer displays separate real-time badges for each station.

The **CSV export** labels intervals as `ground-station-1` or `ground-station-2` in the type column.

5.4 View Controls (Right-Side)

- **Projection:** Toggle 2D Map vs. 3D Globe.
- **Trail (orbits):** Sets how much ground track is drawn, in *orbital periods per satellite* (each satellite's window is computed from its own period, so mixed LEO/GEO scenarios both stay readable). Blank means the full time span; a number N draws a trailing window of N orbits ending at the current time; 0 shows markers only. Applies to both the 2D map and the 3D globe, and affects display only, never the computed tables or CSV.
- **Focus Mode (`↗`):** Expands view to full window.
- **3D camera (mouse and touch):** Drag with one finger or the mouse to rotate; pinch with two fingers or use the scroll wheel to zoom. Rotation is locked while Focus follows a satellite, but zoom still works.
- **3D Controls:**
  - **Reset Camera:** Re-centers the view.
  - **Focus:** Automatically follows Sat 1 or Sat 2.
  - **Terminator:** Shows the day/night boundary on the globe.
  - **LOS link:** Draws the inter-satellite link line whenever line of sight exists.
  - **Inertial:**
    - *ON:* Orbital tracks are fixed in space; Earth rotates underneath (Physically correct).
    - *OFF:* Camera rotates with the Earth (ECEF view).
  - **Elements:** Toggles the HUD showing live orbital elements (Ω, ω).

5.5 Keyboard Shortcuts

The following keyboard shortcuts are available for faster navigation. Shortcuts are disabled when typing in input fields.

Key	Action
`Space`	Play/Pause simulation

Key	Action
`R`	Reset to start
`←` / `→`	Step backward/forward (auto-pauses)
`V`	Toggle 2D/3D view
`I`	Toggle Inertial view (3D only)
`0`	Free camera (no focus)
`1`	Focus on Satellite 1
`2`	Focus on Satellite 2
`E`	Export CSV
`T`	Open TLE import for Satellite 1
`Shift+T`	Open TLE import for Satellite 2

5.6 TLE Import

Each satellite panel includes a collapsible **"Import from TLE"** section that allows you to populate orbital elements from a standard Two-Line Element set.

To use TLE Import:

1. Click **"▶ Import from TLE"** at the bottom of a satellite panel (or press `T` for Sat 1, `Shift+T` for Sat 2).
2. Paste a TLE (2 or 3 lines) into the text area.
3. Optionally check **"Set simulation epoch to TLE epoch"** to also move the simulation start time to the TLE's epoch.
4. Click **"Apply TLE"** to populate the orbital elements. The elements are **anchored at the TLE's own epoch** and propagated (with J2 secular rates) onto the simulation timeline, whatever the simulation start time is. Errors and warnings appear as messages inside the panel rather than as popup dialogs.

Supported TLE format:

```
ISS (ZARYA)              ← Optional name line (ignored)
1 25544U 98067A   24001.50000000  .00016717  00000-0  10270-3 0  9025
2 25544  51.6400 247.4627 0006703 130.5360 229.6106 15.49815571423456
```

What gets extracted:

- Inclination, RAAN, Eccentricity, Argument of Perigee, Mean Anomaly (converted to True Anomaly)
- Mean Motion (converted to Semi-major Axis)
- Epoch (year and day-of-year converted to ISO-8601 UTC)

Understanding TLE Epochs

What is a TLE epoch?

Think of a TLE as a photograph of a satellite's orbit taken at a specific moment. That moment is called the "epoch." The TLE tells you exactly where the satellite was and how it was moving when the photograph was taken. Just like a photograph, a TLE captures a single instant in time.

The mismatch problem (and how the app handles it)

Imagine you have two photographs of friends taken on different days. To place them correctly in the same scene, you would need to account for how each person moved between their photo date and the scene date.

That is exactly what the app does. Suppose:

- Satellite 1's TLE is from **January 15th**
- Satellite 2's TLE is from **January 20th**
- Your simulation starts on **January 15th**

Each satellite is anchored at its *own* TLE epoch, and the propagator carries it forward or backward (orbital motion plus J2 precession) to every point on the simulation timeline. Both satellites are therefore placed consistently, even with mismatched TLE dates. The remaining caveat is model fidelity: over large gaps the J2-only model omits real effects (notably atmospheric drag for low orbits), so accuracy gradually degrades as the gap between the TLE epoch and the simulated times grows. Over five days a LEO satellite completes roughly 75 orbits, so even small unmodeled forces have room to accumulate.

Why positions drift over time

Satellites don't stay still. Between any two dates:

- The satellite completes many orbits, changing its position along the orbital path
- Earth's equatorial bulge causes the orbital plane to slowly rotate (RAAN drift)

- The orientation of the orbit's ellipse also shifts (argument of perigee drift)

These effects accumulate. The app models the orbital motion and the J2 drifts exactly, but not drag or higher-order effects, so a TLE from last week describes today's orbit only approximately.

Normalized vs. raw TLEs

You may notice that some TLEs always show satellites starting at the equator. This happens because certain data providers "normalize" their TLEs so the epoch aligns with the satellite's ascending node (the moment it crosses the equator heading north). This is a processing choice, not a physical requirement.

Raw, unprocessed TLEs from sources like CelesTrak or Space-Track will show satellites at whatever position they occupied when the TLE was generated. Both types are valid; they're just different conventions.

Practical recommendations

- Fresh TLEs are best: keep the gap between each TLE's epoch and the times you simulate to a few days for LEO (drag matters most there); GEO tolerates much larger gaps
- Enable "Set simulation epoch to TLE epoch" for one satellite if you want the timeline to start on real orbital data
- Mismatched TLE epochs between the two satellites are handled automatically by per-satellite anchoring; they no longer need to match each other
- Hand-editing any element of a satellite clears its TLE anchor and reinterprets its elements at the simulation epoch

Note: Dual epoch sync

If you enable "Set simulation epoch to TLE epoch" for *both* satellites and their TLE epochs differ, the simulation *start time* follows the last import ("last write wins"). This is harmless to accuracy: each satellite remains anchored at its own TLE epoch regardless of where the timeline starts. The checkbox only chooses where the timeline begins.

If a TLE's epoch differs from the simulation epoch by more than 7 days, the app shows a non-blocking note in the panel about model fidelity over large gaps; the import proceeds either way. After import, the panel displays "Anchored at TLE epoch" with the epoch for reference (expand the TLE section to see it).

Note: The TLE parser validates that the resulting orbit has perigee above Earth's surface. Invalid TLEs will display an error message.

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6) How The Physics Works

- **Orbit Propagation:** Two-body Keplerian motion with J2 secular perturbations applied to RAAN (Ω) and Argument of Perigee (ω) at every time step.
- **ECI ↔ ECEF:** Standard transformation using Greenwich Mean Sidereal Time (GMST).
- **Horizon/FOV:** Calculated assuming a **Spherical Earth** radius ($R_E = 6378.137$ km).
- **LOS Test:** Geometric intersection test between the satellite-to-satellite segment and the Earth sphere.
- **Per-satellite epochs:** Each satellite propagates from its own anchor epoch (the TLE epoch if imported, otherwise the simulation epoch), with J2 secular rates applied over the elapsed time from that anchor.
- **Contact boundary refinement:** Interval boundaries are first located on the sampled time grid, then refined by bisecting the access condition between the bracketing samples down to about 1 ms, so reported rise/set times and durations are not quantized to the step size. This removes the sampling error; the remaining error budget is set by the model itself (spherical Earth, sea-level stations, no refraction, no drag; see section 8). Boundaries at the very edge of the timeline (no bracketing sample) remain at the sample time.

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7) Typical Workflows

A) Visualize a Sun-Synchronous Orbit

1. In **Simulation Controls**, select **Load Scenario: Sun-Synchronous (J2 Demo)**.
2. Switch to **3D Globe**.
3. Enable **Inertial View** and check **Show Elements**.
4. Scrub the timeline. You will see the orbital plane (Ω) slowly precessing, validating the J2 physics engine.

B) Follow a Satellite in Inertial Space

1. Select **Load Scenario: LEO Formation**.
2. Switch to **3D Globe** and enable **Inertial View**.
3. Set **Focus: Sat 1**.
4. The camera will now "chase" the satellite's position in space while the Earth rotates beneath it.

C) Analyze High-Latitude Access (Molniya)

1. Select **Load Scenario: Molniya**.
2. Observe the 2D Map. Notice how the satellite spends the majority of its time (the "apogee dwell") over the northern hemisphere, maximizing contact with the high-latitude Ground Station.

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8) Simulation Fidelity & Limitations

This tool is designed for educational demonstration and first-order analysis.

- **J2 Dynamics:** The model includes J2 secular perturbations but **does not** include atmospheric drag, solar radiation pressure, or third-body gravity (Moon/Sun).
- **Spherical Ground Model:** All ground calculations assume a perfect sphere.
  - *Implication:* Ground tracks and footprints may have slight deviations compared to a WGS-84 ellipsoid model, particularly at high latitudes.
  - *Note:* While visually sufficient for demos, mission-critical analysis should use high-fidelity software (STK, GMAT).
- **TLE interpretation:** TLE elements are treated as osculating Keplerian elements in the app's frame. The TEME reference frame and the Brouwer/Kozai mean-motion convention behind real TLEs are not modeled, which introduces small errors (a few kilometers in semi-major axis for LEO). Appropriate for education and demonstration; use SGP4-based tools when that matters.
- **Ground station altitude:** Stations sit at sea level on the spherical Earth; site elevation is not an input. Access boundary times shift accordingly, most noticeably against low elevation masks.
- **Atmospheric refraction:** Not modeled. Real radio and optical paths bend near the horizon, so geometric rise/set times differ from observed ones by seconds to tens of seconds at low masks.

- **Net effect on contact times:** The bisection refinement means reported boundaries are exact for this model, but real-world pass-time accuracy is typically seconds to tens of seconds, dominated by the simplifications above and by TLE age.

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9) Troubleshooting

- **Blank Screen / No Tracks:** Ensure your browser supports WebGL. Check that the Epoch is in a valid ISO-8601 format (e.g., `2025-01-01T12:00:00Z`).
- **Can't Rotate Globe:** Check if **Focus** is set to a satellite. Set it to **None** to regain free manual control. On touch devices, rotate with one finger and pinch with two.
- **"Custom" appears in Scenario box:** This is normal. If you edit any physics parameter manually, the app switches the label to "Custom" to indicate the settings no longer match a preset.

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10) Release Notes

v1.4.0

- **Physics:** Contact interval boundaries are refined by bisection between time samples (sub-second precision within the model) instead of being quantized to the step size. Tables display durations at 0.1 s resolution; CSV timestamps carry milliseconds.
- **Physics:** Per-satellite element epochs. TLE imports anchor each satellite at its own TLE epoch; two TLEs with different epochs are both honored. Hand edits or scenario loads clear the anchor.
- **Feature:** Trail control unified and moved to the view header. Units are orbital periods per satellite; blank = full span; applies to both 2D and 3D views. Fixes the old bug where a trail of 0 silently became 30 minutes.
- **Mobile:** 3D camera uses Pointer Events (one finger rotates, two fingers pinch-zoom); canvases size responsively (map 2:1, globe 1:1, capped at desktop sizes); 3D control pills and checkboxes render correctly on tablets; scrubbing no longer jitters the page.
- **UI:** The ten Yes/No visibility dropdowns are now one-tap checkboxes with properly associated labels; all sidebar labels gained programmatic association. TLE errors and warnings appear inline in the panel instead of

popup dialogs; the epoch-gap warning is now a non-blocking note (threshold 7 days).
- **UI:** Buttons restyled with white text on palette-derived fills; focus ring recolored to the theme and switched to keyboard-only (focus-visible). Export button relabeled "Export CSV"; playback control relabeled "Speed (steps/frame)"; 3D pill labels shortened (Terminator, LOS link, Inertial, Elements). Map style control removed (textured globe is always used).
- **Fixes:** Applying a TLE now correctly flips the scenario dropdown to "Custom"; interval tables use a scroll wrapper so columns size naturally; save-artifact residue removed from the markup.
- **Legal:** Footer and LICENSE years updated to 2025-2026; contradictory "All rights reserved" phrasing removed alongside the MIT license statement.

v1.3.0

- **Feature:** Added **Ground Station 2**. The application now supports two independent ground stations, each with its own coordinates, elevation mask, and access calculations.
- **UI:** New "Ground Station 2" panel (disabled by default). GS1 shown as white triangle, GS2 as green triangle on both 2D and 3D views.
- **UI:** Access intervals table now includes a "Station" column and merges both stations' intervals chronologically.
- **UI:** Separate footer badges for GS1 and GS2 real-time access status.
- **Export:** CSV now labels intervals as `ground-station-1` or `ground-station-2`.
- **Code Quality:** Removed stray Evernote/Skitch CSS artifacts (176 lines) and browser-injected comment from file.
- **Documentation:** Updated User Manual with new Ground Station Panels section (5.3).

v1.2.0

- **Feature:** Added **TLE Import** to parse Two-Line Element sets and populate orbital elements.
- **UI:** Collapsible "Import from TLE" section in each satellite panel with epoch sync option.
- **Keyboard:** Added `T` (Sat 1) and `Shift+T` (Sat 2) shortcuts for TLE import.
- **Documentation:** Updated User Manual with new TLE Import section (5.6).

v1.1.2

- **Feature:** Added **Keyboard Shortcuts** for improved workflow (Space, R, arrow keys, V, I, 0/1/2, E).
- **Documentation:** Updated User Manual with new Keyboard Shortcuts section (5.5).

v1.1.1

- **Fix:** Corrected negative longitude and angle wrapping functions.
- **Fix:** Added numerical tolerance for 90° elevation mask edge case.
- **Fix:** Removed dead UI references and unified terminator toggle handling.
- **Code Quality:** Renamed internal True Anomaly variables from `M` to `nu` for semantic correctness.

v1.1.0

- **Feature:** Added **Scenario Presets**. Users can now instantly load LEO, SSO, GEO, and Molniya configurations via a dropdown menu.
- **UI:** Added logic to auto-detect manual edits and switch scenario state to "Custom."
- **Core:** Validated J2 propagator against standard SSO parameters.
- **Documentation:** Updated User Manual to reflect the new workflow.

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